Impact of Harmala (Peganum harmala, L.) Seeds on Biological, Biochemical and Histopathological Alternations of Alloxan-Induced Diabetic Rats

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Abstract

The goal of the current study was to ascertain the effects of Harmala seed powder and extract rats that had been given alloxan to induce diabetes. Thirty-six white randoms weighing between 140 and 150 g were randomly divided into two main groups of six. Group I was the randomized control group, while Group II received a single intra-cage injection of the diabetic medication alloxan (150 mg/kg b.wt.). Next, the diabetic rats in Group II were randomly divided into seven subgroups: For four weeks, groups 3-6 received the ordinary food plus 2.5 and 5% Harmala seed powder, as well as 250 and 500 mg/kg. Group 2, the diabetic group serving as the positive control, received only the standard diet. Twenty-eight days after the trial started samples were analyzed.

The obtained data showed that the studied plants ($P \leq 0.05$) decreased serum sugar, Low density lipoprotein (LDL-c), and High-density lipoprotein (HDL-c) ($101.70^{e} \pm 3.12$, $29.71^{e} \pm 1.17$, $45.90^{b} \pm 1.34$ when compared to the control positive group. The plants under study also improved blood parameters, liver, and kidney function. Liver enzymes Alanine aminotransferase (ALT), Aspartate aminotransferase (AST), Alkaline phosphatase (ALP) ($90.73^{c} \pm 1$. $153.61^{c} \pm 2.5542$, $154.52^{c} \pm 2.61$). According to our research, the powdered Harmala seeds and their extract include several potent phytochemical substances that may help reduce side effects and regulate diabetic rats.

Key words: Total cholesterol, serum glucose, liver function, kidney function, Alanine aminotransferase, Aspartate aminotransferase, Alkaline phosphatase

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1. INTRODUCTION

Diabetes mellitus is a group of metabolic disorders characterized by hyperglycemia, with disturbances of carbohydrate, fat and protein metabolism resulting from defects in insulin, insulin action, or both **(Nayak *et al.,* 2011)**. The diminution in viability has motivated trials to supersede foregoing by whole pancreas transplant. The transplanted pancreatic islets were either segregated from organ givers or gained from stem cells. Each of these approaches has constructed important progress upon the past 2 era. Despite that, billions of patients are not satisfied with these methodologies for motives of cost and givers organ availability **(Kotb *et al.,* 2024)**. The long-term metabolic illness regarded as diabetes mellitus, which is demonstrated by using the body's lack of ability to create or use insulin, has an influence on a person's social, physical, and mental health **(Mustapha *et al.,* 2023)**. In the body, the pancreas generates insulin, which aids in transferring glucose from the circulation to the cell tissues, the place where it is similarly metabolized and used as fuel for healthy metabolic processes. **(Srinivasan *et al.,* 2019)**. By 2030, the World Health Organization predicted that there would be about 366 million human beings global with diabetic, with men and women aged round 45 and 64 years experiencing the largest growth. Thus, it is essential to decrease **(Leslie *et al.,* 2016)**.Diet, exercise and chemical glucose lowering medicine are used for blood glucose control but despite the use of multiple oral medications many patients suffer from high blood glucose and most patients do not accept insulin injection **(Degli-Esposti *et al.,* 2013)**. Uncontrolled diabetes mellitus is associated with serious micro and macrovascular complications **(Bash *et al.,* 2008)**.

A multiplicity of herbs besides medicinal plants with amalgams sterilized from them reassessed as an alternative healing of diabetes throughout the globe. To the same degree they possibly will specify a base of neoteric against -diabetic analogues through influential efficacy. They are lacking the undisciplined side effects associated with conventional drug and high risk of consequent malfunction **(Li *et al., 2017)***. Harmala (Peganum *harmala,* L.) is a desolate growing blossoming plant, unearthed richly in MiddIe East plus North Africa. It is an adaptable medicinal seedling giving

copious pharmacological activities attributed to the active alkaloids harmala **(Bensalem *et al.,* 2017)**. Harmala is a member of the family *Zygophyllaceae*, commonly known as Harmel grows spontaneously in semi-arid and pre-desertic regions of south-east Morocco and distributed in north Africa and the middle east **(EL-Bahri and Chemli, 1991)**. Harmala seeds has been traditionally used to treat diabetes in folk medicine of some parts of the world **(Bnouham *et al.,* 2002)**. This effect of Harmala has been pharmacologically confirmed in several studies, one of which showed that the plant would lose its hypoglycemic activity at high doses instead of increasing it **(Nafisi *et al.,* 2011)**. The biochemical possessions of the unmitigated alkaloids extracted from Harmel are registered concerning many studies which exposed its safety and efficiency to regenerative remedy. Consequently, harmala are an exploration motivation in the medical and pharmaceutical practices **(Shao *et al.,* 2017)**. The anti-diabetic effects of Harmala seeds possibly related to its antioxidant properties and and the agonist/antagonistic effect proceeding with responsible receptors. the hypoglycemia outcome of the seed isolate of harmala start from the augmented insu1in secretion and enhanced proliferation of pancreatic in diabetic rats **(Moin *et al.,* 2021)**. Harmine is competent found in Harmala seeds, it is proliferation initiation, multiplicate islet reservoir and improve glycemic control ***(Shaw et al., 2016).*** These observations suggest that harmine analogs possibly will have distinctive therapeutic promising for human diabetes therapy. Enhancing the potency and beta cell specificity of these compounds are principal future challenges (**Mohamady *et al.,* 2024)**. Therefore, was designed to determine Harmala seeds effects on alloxan induced diabetes levels in males Albino rats.

Materials and Methods

Materials

Harmel seeds

In March 2024, herbalists in Cairo City, Cairo Governorate, old the Harmala seeds (*Peganum harmala*, L.).

Animals (rats)

Thirty-six male Albino rats weighing between 140 and 150 grams were provided by the Ministry of Health's Vaccine and Immunity Organization at Helwan Farm in Cairo, Egypt.

The chemical and kits

Al-Gomhoria Company for Trading Drug, Chemical and Medical Instruments, Cairo, Egypt, provided Alloxan and the kits for the glucose, lipid fraction, liver, and kidney biomarkers used in this study.

Methods*:*

The preparation of Harmel seeds powder

According to **(AOAC, 2005)**, the Harmala seeds were ground into a powder using a German-made Broun high-speed blender, then sealed in black glass bottles until additional processing.

Preparation of Harmala seeds extract

In a 5-liter glass beaker, the seed powder was macerated in 100% ethanol for seven days, occasionally shaking and stirring. Whatman no 8-filter paper was used to filter the macerated material after seven days. A rotary evaporator operating at 40°C was used to evaporate the solvent. as reported by **Muhammad *et al.*, (2024)**.

Quantification of Harmala compounds using HPLC

The phenolic components found in Harmala extracts were chemically analyzed using HPLC using a modified approach of **Kartal *et al.*, (2003)**. Extract samples were filtered with a 0.45 µm PTFE syringe tip filter and a 20µL sample loop. To identify and quantify phenolic components in harmala, an RP-HPLC system with a Shimadzu PDA detection, reverse phase (RP C-18) column (Interstsil ODS-5 µm x 4.6 mm x 250 mm), quaternary pump, and online vacuum degasser was utilized. The flow rate was set at 0.8 mL/min at 37°C. For this study, a gradient of two mobile phases was used. Solvent A was made up of water, acetonitrile, and trifluoroacetic acid in the proportions of 90:10:0.1% (v/v), while solvent B was made up of 10:90:0.06% (v/v). Elution was done using the following

gradient: 0-10 min: 10-35% B; 10-20 minutes: 35-42% B; and 20-30 minutes: 42-100% B. Following each run, the system was reconditioned for 10 minutes before analyzing the next sample.

Basal diet

The created basal diet that was developed with the **AIN (1993)** formula: 10% protein, 5% cellulose, 10% corn oil and vitamin blend, 1% minerals, and 69.5% maize starch. Additionally,

Inducing damage to pancreatic (diabetic) in rats

In healthy, normal rats, an injection of alloxan (150 mg per kg of body weight) resulted in long-term damage to the beta cells of the pancreas **(Zhao et al., 1987)**. Rats with diabetes were given injections of alloxan, and a week later, fasting blood samples were obtained to measure their fasting serum glucose levels, which were 200 mg/dl **(NDDG, 1979)**.

Experimental animals

Research IRB No. 1320240113 was approved by the Faculty of Specific Education's Research Ethics Committee at Assiut University. Thirty-six adult male white albino rats of the "Sprague Dawley" weighing 140-150g, were utilized in this experiment. All rats were given a for seven days to help them adapt. Following this period of adaptation, the rats were split up into six groups, with six rats in each: **Group 1:** As a negative control group, rats have been given a standard diet only. **Group 2:** As a control positive group, hyperglycemic rats were given a standard diet. **Group 3:** Hyperglycemic rats were fed a basal diet and recieved2.5% Harmel seeds of the weight of the diet. **Group 4:** After being fed a basal diet, hyperglycemic rats were given 5.0% Harmel seeds of their diet weight. **Group 5:** 250 mg/kg Harmel seed extract of the diet's weight was administered to hyperglycemic rats after they were fed a basal diet. **Group 6:** 500 mg/kg Harmel seed extract of the diet's weight was administered to hyperglycemic rats after they were fed a basal diet.

Collection of blood and organs

Blood samples from a portal vein have been collected in sterile, dry centrifuge tubes. The samples were centrifuged for 10 minutes at 4000 rpm to extract the serum. For analysis, the serum was first stored at -20°C using the Schermer (1967) procedure. For histological analysis, the pancreas was previously removed concurrently, cleaned with saline solution, dried using filter paper, weighed, and preserved in 10% formalin liquid ***(Drury et al., 1980).***

Biochemical analysis

A technique of **(Barham and Trinder, 1972)** was used to estimate serum glucose. The serum insulin level using **Temple *et al.*, (1992)**. The methods were used to measure triglycerides. The **Allian (1974)** method was used to determine total cholesterol. The **Lopez (1977)** approach was used to calculate the HDL-c values. Both VLDL-c and LDL-c were calculated using the following equation: VLDL-c =Triglyceride/5. According to **Lee and Nieman (1996)**, LDL-c is equal to (T.C. - HDL-c) - VLDL-c. The methods of the **IFCC (1983)** for estimating alkaline phosphatase (ALP) concentration, **Young (1975)** for estimating alanine amino transferase (ALT) levels, and **Henry *et al.*, (1974)** for estimating aspartate amino transferase (AST) levels were used. On the other hand, the methods of **(Patton and Crouch, 1977), (Zhao *et al.*, 2009) and (Schirmeister, 1964)** were used to utilize for measuring serum creatinine, urea, and uric acid. The method described by **International *et al.*, (2004)** was used to determine the concentration of hemoglobin (HGB), red blood cell count (RBC, s), white blood cell count (WBC, s), and platelet count (PLT).

Statistical

In order to identify a significant primary factor, a fully randomized factorial design was employed to investigate the data, and the Student-Newman-Keuls test was performed to separate the abilities. The Costate Program has determined that there is a significant difference between treatments ($P \leq 0.05$). Originally, one-way ANOVA was employed to take biological impacts into account **(SAS, 1988)**.

Results and Discussion

Table 1 displays the results of the HPLC method used to quantify Harmala seeds. It is clear to notice that, harmol, chebulic acid, and harmaline had the greatest amounts of phenolic compounds in Harmala seeds, with respective values of 852.11, 534.30, and 463.13 µg/g. Quinic acid, peganine, and kaempferol had the lowest values of Harmala seeds, with respective values of 170.14, 200.35, and 272.36 µg/g. However, under these conditions, neither gallic acid nor ferulic acid were found. These findings are consistent with those of (**Kamran *et al.*, 2024)**, who found that 17 distinct bioactive compounds from phenolic acids, flavonoids, and their conjugates with carbohydrates components of plant cells are present in Harmala seeds. According to a recent study, Harmala can be used to create functional foods and in medication formulations to treat a wide range of illnesses.

Additionally, the organic extract of Harmala seeds contained 874, 488, and 380 mg/ml of harmol, harmaline, and harmine, respectively, which were identified and measured. According to) Li ***et al.*, 2018)**, these discovered chemicals demonstrated physiologically active molecules from this historically significant medicinal plant.

Table (1): Quantification of Harmala seeds fractions by HPLC

Phenolic substances	Concentration µg/mg
Harmine	375.11
Harmaline	463.13
Harmol	852.64
Peganine	200.35
Quinic acid	170.14
Chlorogenic acid	291.25
Catechin	286.14
Caffeic acid	277.32
Chebulic acid	534.30
Kaempferol	272.36
Gallic acid	ND
Ferulic acid	ND

ND= Not detectable.

Influence of Harmala seeds powder and its extract on diabetic rats' blood glucose levels

Table (2) shows that Harmala seeds powder and its extract influenced the glucose and insulin levels of diabetic rats. The findings revealed significant variations in sugar levels between the two control groups, with the negative control group having the lowest value. The corresponding averages were 270.15 and 91.85 mg/dl. The groups who were given 2.5% Harmala seeds powder had the highest glucose levels, whereas the group administered 500mg/kg Harmala seeds extract had the lowest value with significant variations.

There is a significant difference in insulin levels between the two control groups, with the negative control group recording the highest amount. G1 averaged 34.02 mIU/L, while G2 averaged 16.44 mIU/L. Regarding the diabetic groups, data revealed that the groups given 2.5% Harmala seeds powder had the lowest insulin levels, while the group administrated 500mg/kg Harmala seeds extract group had the highest value with a considerable variation. The average values were 30.05 and 21.09 mIU/L. These findings supported by (**Hamden *et al.*, 2009)**'s discovery that serum glucose levels were considerably lower in all experimental groups received varying doses of the organic extract of harmala seed (100, 200, and 400 mg/kg) compared to the control.

Harmala seeds may have antidiabetic characteristics due to their antioxidant capabilities, enzymatic inhibition, or by producing agonist/antagonistic actions on key receptors **(Komeili *et al.*, 2016)**. Furthermore, they indicated that the hypoglycemic effect of Harmala's organic seeds extract could be attributed to increased insulin secretion by diabetic rats' surviving pancreatic beta cells. Thus, Harmala's hypoglycemic activity can be linked to glucose consumption or absorption rather than the pancreas **(Ae Park *et al.*, 2006)**.

Table (2) Influence of Harmala seeds powder and its extract on glucose and insulin level of diabetic rats

Parameters / Groups	Glucose mg/dl	Insulin mIU/L
G_1 C (-)	91.85^{f}±2.17	34.02^{a}±1.24
G_2 C (+)	270.15^{a}±4.35	16.44^{f}±1.05
G3 (2.5% Harmala seeds powder)	150.00^{b}±3.46	21.09^{e}±1.23
G4 (5% Harmala seeds powder)	136.34^{c}±3.53	24.18^{d}±1.25
G5 (250 mg/kg Harmala seeds extract)	112.85^{d}±3.24	27.22^{c}±1.23
G6 (500 mg/kg Harmala seeds extract)	101.70^{e}±3.12	30.05^{b}±1.24
LSD (P≤ 0.05)	**4.151**	**1.034**

Each value represents the mean ± SD of three replicates. Means in the same column with different letters are significantly different (P<0.05).

Influence of Harmala seeds powder and its extract on liver enzymes of diabetic rats

Table (3) shows powder and its extract altered the ranges of liver activities (ALP, AST, and ALT) in diabetic rats. The data collected revealed that the group with a positive control had the greater amount of ALT value, while the group with a negative control had the lowest value, with significant differences. The average amounts were 125.55 and 60.28 U/L. With significant variation, the 500 mg/kg Harmala seeds extract had the lowest ALT ranges among the treatment groups, while the 2.5% Harmala seeds powder had the greatest value.

In the case of serum AST, it was previously shown that the group with control positive had greater AST enzyme ranges, while the group with control negative had the lowest value, with significant variances. The common averages are 253.15 and 120.50 U/L. The 2.5% Harmala seeds powder group had the highest value of AST ranges among the treated groups, while the 500 mg/kg Harmala seeds extract group had the lowest value, with significant variations, which were 162.35 and 137.35 U/L.

Furthermore, the findings revealed that the control positive group had the highest levels of ALP enzyme, while the negative control group had the lowest levels, with statistical differences; corresponding mean values were 261.30 and 130.85 U/L. The 2.5% Harmala seeds powder group had the highest ALP concentrations, whereas the 500 mg/kg Harmala seeds extract group had lower values with statistical variations, averaging were 178.42 and 140.50 U/L, respectively. These findings supported the observations of (**Zaker *et al.*, 2008)**, who reported that Harmala seeds contain active chemicals such as vitamins A, B, C, E, and K, as well as high mineral content such as salt, iron, and calcium. These chemicals exhibit high antioxidant and hepatoprotective properties.

Qi *et al.*, (2010) discovered that flavonoids chemicals found in Harmala seeds have hepatoprotective properties and are currently utilized in therapeutic hepatic treatments (**Qi *et al.*, 2010)** .

Table (3) Influence of Harmala powder and its extract on liver enzymes of diabetic rats

Parameters / Groups	ALT U/L	AST U/L	ALP U/L
Parameters Groups	**ALT U/L**	**AST U/L**	**ALP U/L**
G_1 C (-)	60.28^{f}±1.10	120.50^{e}±2.11	130.85^{e}±1.87
G_2 C (+)	125.55^{a}±2.53	253.15^{a}±3.41	261.30^{a}±3.52
G3 (2.5% Harmala seeds powder)	100.10^{b}±2.24	162.35^{b}±2.60	178.42^{b}±2.40
G4 (5% Harmala seeds powder)	90.73^{c}±1.42	153.61^{c}±2.55	154.52^{c}±2.61
G5 (250 mg/kg Harmala seeds extract)	82.61^{d}±1.38	143.45^{d}±2.63	152.60^{c}±2.57
G6 (500 mg/kg Harmala seeds extract)	74.90^{e}±1.21	137.35^{d}±2.42	140.50^{d}±2.36
LSD (P≤ 0.05)	**2.310**	**2.413**	**2.501**

ALT=Alanine aminotransferase. AST= Aspartate aminotransferase. ALP=Alkaline phosphatase. Each value represents the mean ± SD of three replicates. Means in the same column with different letters are significantly different (P<0.05).

Influence of Harmala seeds powder and its extract on total cholesterol and triglycerides of diabetic rats

Table (4) shows the effects of Harmala seed powder and extract on total cholesterol (TC) and triglyceride (TG) levels in diabetic rats. The results confirmed that, with significant differences, the control positive group had the highest value of cholesterol value, while the control negative group had the lowest. The suggested averages were 185.20 and 81.35 mg/dl. For the treatment groups, 2.5% Harmala seeds powder showed the highest serum cholesterol levels, while 500 mg/kg Harmala seeds extract showed the lowest, with statistical differences. The recommended values were 147.13 and 92.70 mg/dl.

In terms of triglycerides, it was previously clear that the group with control positive had the widest range, whereas the group with control negative had the smallest value, with statistically significant differences. The relative averages were 162.43 and 70.10 mg/dl. The 2.5% Harmala seeds powder group had the highest value of blood triglyceride ranges among treated groups, while the 500 mg/kg Harmala seeds extract group had a lower value with statistically significant changes. The relative averages were 126.43 and 85.45 mg/dl. These findings are supported by **(Kalhora *et al.*, 2015)**, who discovered that different levels of harmala seeds methanolic extract reduced total cholesterol by 100-400 mg/kg harmala. This was due to the inhibition of 3-hydroxy3methyl-glutaryl-CoA (HMGCoA) reductase by various alkaloids harmine, harmaline, and harmol found in harmala seeds. These alkaloids have also been shown to have a diabetic impact **(Singh *et al.*, 2008)**.

Additionally, harmala seeds and oil extract considerably reduced total cholesterol, total lipids, and triglyceride concentrations, resulting in a reduction in alpha amylase activity.

Table (4) Influence of Harmala seeds powder and its extract on serum triglycerides, and total cholesterol of diabetic rats

Parameters / Groups	Total cholesterol mg/dl	Triglycerides mg/dl
G_1 C (-)	81.35^{f}±1.25	70.10^{f}±1.33
G_2 C (+)	185.20^{a}±1.73	162.43^{a}±1.80
G3 (2.5% Harmala seeds powder)	147.13^{b}±1.45	126.40^{b}±1.53
G4 (5% Harmala seeds powder)	129.41^{c}±1.60	102.17^{c}±1.46
G5 (250 mg/kg Harmala seeds extract)	124.25^{d}±1.37	94.32^{d}±1.38
G6 (500 mg/kg Harmala seeds extract)	92.70^{e}±1.41	88.45^{e}±1.27
LSD (P≤ 0.05)	**2.455**	**2.172**

Each value represents the mean ± SD of three replicates. Means in the same column with different letters are significantly different (P<0.05).

Influence of Harmala seeds and its extract on the lipoprotein cholesterol fraction of rats suffering from diabetes

Table (5) shows how Harmala seed powder and extract impact the values of lipid fractions HDL-c, LDL-c, and VLDL-c in diabetic rats. The control negative group had the highest ranges of HDL-c, while the control positive group had the lowest levels, with statistically significant variations, the average readings being 60.33 and 27.89 mg/dl, respectively. In contrast, the 500 mg/kg Harmala seeds extract group had the highest HDL-c ranges of the treatment groups, while the 2.5% Harmala seeds powder group had the lowest values with statistically significant changes. The corresponding average levels were 45.90 and 31.54 mg/dL.

Furthermore, the results confirmed that the group with control positive had the highest LDL-c readings, while the group with control negative had the lowest, with statistically significant differences. The relative averages had been 124.82 and 17.00 mg/dL. For the treatment groups, 2% Harmala seeds powder had the highest serum LDL-c level, while 500 mg/kg Harmala seeds extract had the lowest value, with

statistically significant differences. The averages were 90.31 and 29.71 mg/dL.

In terms of VLDL-c, the group with control positive had the highest value, while the group with control negative had the lowest value; these differences were significant. There were two different mean levels: 32.49 and 14.02 mg/dL. For the treatment groups, 2.5% Harmala seeds powder had the highest value of VLDL-c levels, while 500 mg/kg Harmala seeds extract had the lowest value with statistically significant changes; the average reading was 25.28 and 17.09 mg/dl, respectively. These findings supported (**Berrougui *et al.*, 2006)** observation that Harmala's antihyperlipidemic action could be attributed to the presence of alkaloids such as harmaline and harmine, which have significant free radical scavenging potential and inhibit lipid peroxidation.

Likewise, oral administration of Harmala seed extract resulted in a significant drop in total cholesterol, triglycerides, and LDL-C levels, as well as a significant increase in HDL-C levels when contrasted with untreated obese-diabetic rats. The antioxidant activity of Harmala may be attributed to the presence of phenolic components such as flavonoids and tannins in the methanolic extract **(Hayet *et al.*, 2010)**.

Furthermore, administering an ethanol extract of Peganum harmala seeds for 14 days resulted in a dose-dependent decrease in blood total cholesterol, triglycerides, and LDL-c levels and simultaneously boosting HDL content. Furthermore, the ethanol extract of Harmala seeds has the potential to decrease blood sugar levels in all three animal studies and improve hyperlipidemia in animals, owing to the extract's high total phenolic as well as total flavonoids content, as well as its powerful antioxidant properties (**Muhammed *et al.*, 2024)**.

Table (5): Effect of Harmala seeds powder and its extract on lipid profile of diabetic rats

Parameters / Groups	HDL-C mg/dl	LDL-c mg/dl	VLDL-c mg/dl
G_1 C (-)	50.33^{a}±1.54	17.00^{f}±0.15	14.02^{f}±0.25
G_2 C (+)	27.89^{f}±1.15	124.82^{a}±1.86	32.49^{a}±0.61
G3 (2.5% Harmala seeds powder)	31.54^{e}±1.33	90.31^{b}±1.40	25.28^{b}±0.53
G4 (5% Harmala seeds powder)	39.60^{d}±1.42	69.38^{c}±1.35	20.43^{c}±0.40
G5 (250 mg/kg Harmala seeds extract)	41.99^{c}±1.01	63.40^{d}±1.60	18.86^{d}±0.43
G6 (500 mg/kg Harmala seeds extract)	45.90^{b}±1.34	29.71^{e}±1.17	17.09^{e}±0.26
LSD (P≤0.05)	**2.262**	**2.613**	**1.116**

LDL-C=Low-density lipoprotein. VLDL-C. Very low-density lipoprotein. HDL-C= High-density lipoprotein. Each value represents the mean ± SD of three replicates. Means in the same column with different letters are significantly different (P<0.05).

Influences of Harmala seeds powder, and its extract on the renal functions of diabetic rats

Table (6) shows the effect of Harmala seeds powder and extract on the levels of kidney biomarkers in diabetic rats. The group with positive control had the highest urea levels, whereas the group with negative control had the lowest, with statistically significant differences. The average values were 45.90 and 19.50 mg/dL. The 2.5% group of Harmala seeds powder had the highest urea levels across hyperglycemic groups, whereas the 500mg/kg Harmala seeds extract group had the lowest, with statistically significant differences. The relative mean values were 40.34 and 28.38 mg/dl.

The data also revealed that, with statistically significant variances, the effective group with control positive had the highest uric acid levels, while the group with control negative had the lowest, at 6.32 and 2.95 mg/dl on average, respectively. On the other hand, 2.5% of Harmala seeds powder had the highest percentage of blood uric acid ranges in the treatment groups, but 500mg/kg Harmala seeds extract had the lowest amount with statistically significant changes, which were 5.39 and 3.92 mg/dl on average, respectively.

The group with control positive had the highest creatinine levels, whereas the group with control negative had the lowest values with significant (P≤0.05) variances. The relative mean levels were 1.22 and 0.61 mg/dL. In contrast, the 500mg/kg Harmala seeds extract group had the lowest serum creatinine ranges for hyperglycemic groups, whereas 2.5% Harmala seeds powder had the highest value with substantial variance. On average, the concentrations were 1.00 and 0.70 mg/dl. Our findings are consistent with those of (**Abd El Baky *et al.*, 2016)**, who stated that Harmala seeds, its oil and medication dramatically lowered blood renal functions such as urea, uric acid, and creatinine levels in diabetic rats compared to the diabetic control group.

Similar findings were reported by (**Arulselvan *et al.*, 2006)**, who discovered that streptozotocin-induced diabetic rats had significantly higher serum urea, uric acid, and creatinine levels than normal control and treated (Harmala seeds diabetic rats).
(Long *et al.*, 2005) discovered that flavonoids compounds can reduce nephrotoxicity, enhance kidney function, and increase kidney primary epithelial tubular cell regeneration. Aside from that, alkaloids compounds have protecine impacts on improving kidney biomarkers.

Table (6): Influence of Harmala seeds powder and its extract on renal functions of hyperglycemic rats

Parameters / Groups	Urea mg/dl	Uric acid mg/dl	Creatinine mg/dl
G_1 C (-)	19.50^{f}±1.02	2.95^{bc}±0.17	0.61^{ab}±0.10
G_2 C (+)	45.90^{a}±1.57	6.32^{a}±0.61	1.22^{a}±0.31
G3 (2.5% Harmala seeds powder)	40.34^{b}±1.40	5.39^{a}±0.50	1.00^{a}±0.37
G4 (5% Harmala seeds powder)	34.89^{c}±1.32	4.99^{ed}±0.46	0.90^{a}±0.39
G5 (250 mg/kg Harmala seeds extract)	32.89^{d}±1.51	4.62^{b}±0.34	0.81^{a}±0.25
G6 (500 mg/kg Harmala seeds extract)	28.38^{e}±1.36	3.92^{bc}±0.27	0.70^{a}±0.11
LSD (P≤0.05)	**1.132**	**1.035**	**0.340**

Each value represents the mean ± SD of three replicates. Means in the same column with different letters are significantly different (P<0.05).

Influences of Harmala seeds powder, and its extract on the blood parameters of diabetic rats

The data in Table (7) illustrates the effect of Harmala seeds powder and extract on diabetic rats' blood profiles (WBCs and RBCs, HGB, and PLTs). The positive control group had significantly lower RBC levels than the negative control group ($P < 0.05$). The average values were 9.87 and 5.83103/mm3, respectively. The group that got 500 mg/kg Harmala seeds extract had the greatest total RBC levels, while the group fed 2.5% Harmala seeds powder had the lowest value ($P \leq 0.05$). The average values were 7.59 and 9.49 103/mm3, respectively.

The positive control group had significantly greater WBC counts compared to the negative control group ($P < 0.05$). The average values were 4.83 and 2.89 106/mm3, respectively. The group given 500 mg/kg Harmala seeds extract had the highest total WBC counts, while the group fed 2.5% Harmala seeds powder had the lowest ($P \leq 0.05$). The average values were 3.92 and 3.22 106/mm3, respectively.

The positive control group had significantly lower HGB levels than the negative control group ($P < 0.05$). The average readings were 10.39 and 16.11 g/dl, respectively. The group that received 500 mg/kg Harmala seeds extract had the highest HGB levels, while the group that consumed 2.5% Harmala seeds powder had the lowest value ($P \leq 0.05$). The average readings were 15.21 and 12.26 g/dl, respectively.

The positive control group had significantly greater platelet levels than the negative control group ($P \leq 0.05$). The average values were 793.0 and 251.50 106/mm3, respectively. Platelet levels were highest in the 2.5% Harmala seeds powder group, but lowest in the 500 mg/kg Harmala seeds extract group ($P < 0.05$). The average values were 514.50 and 306.0 106/mm3, respectively. These findings are consistent with those published by(**Mahdeb *et al.*, 2020)**, who found a substantial rise in red blood cells, hematocrit, and hemoglobin in animals treated with Harmala seeds on the fifth day as compared to those treated on the first day and control animals. WBCs, and PLTs, counts were dramatically restored in the plant extract

(Harmala seeds) treated groups when compared to non-treated arthritic rats. The most common hallmark of RA is anemia, which is caused by a decrease in red blood cells, and hemoglobin.

Furthermore, rats treated with Harmala seeds levels 400, and 600 mg/kg had significantly higher RBCs, than the treated control group. The red blood cells, in Harmala seeds 400, and 600 mg/kg treated rats were considerably higher than those in control rats **(Akhtar *et al.*, 2021)**.

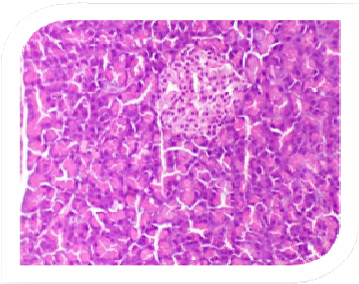
Table (7): Effect of different levels of Harmala seeds powder and its extract on blood parameters in diabetic rats

Groups	Parameters			
	RBC_S (10^6/mm3)	**WBC_S (10^3/mm3)**	**Hemoglobin g/dl**	**Platelet 10^6/mm^3**
G_1 C (-)	2.89^b±0.11	5.83^c±0.16	16.11^a±0.20	251.5^f±3.20
G_2 C (+)	4.83^a±0.50	9.87^a±0.50	10.39^d±0.20	793.0^a±7.20
G3 (2.5% Harmala seeds powder)	3.22^b±0.20	9.49^a±0.61	12.26^c±0.20	514.5^b±5.20
G4 (5% Harmala seeds powder)	3.68^b±0.30	9.04^a±0.51	13.33^c±0.20	446.0^c±5.20
G5 (250 mg/kg Harmala seeds extract)	3.89^a±0.42	7.91^b±0.43	14.88^b±0.20	329.5^d±4.20
G6 (500 mg/kg Harmala seeds extract)	3.92^a±0.40	7.59^b±0.38	15.21^b±0.20	306.0^e±3.30
LSD (P≤0.05)	**1.072**	**1.250**	**1.160**	**5.370**

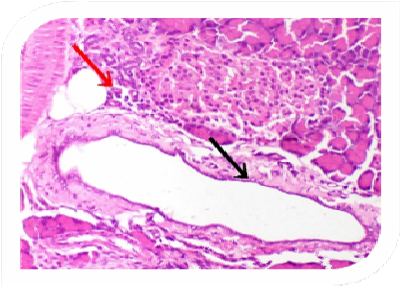
Red blood cells= (RBCs). Wed blood cells= (WBCs). Hemoglobin= (HGB). Platelets =(PLTs). Each value is represented as Values are expressed as mean ± SD; in the same raw with different letters are significantly different (P<0.05).

The pancreas' histopathology:

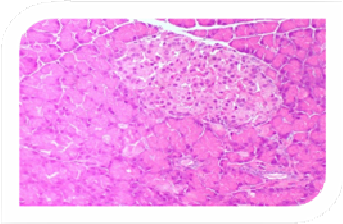
These findings are consistent with those of (**Safamanesh *et al.*, 2021)**, who found that Harmala seed extract and harmine were both beneficial in inhibiting the process of apoptosis and biochemical factors in pancreatic histology. Furthermore, staining results revealed that the seed extract was more effective than harmine in reducing tissue damage.



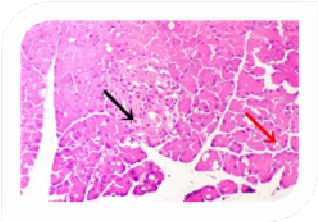
Picture (1): Photo of pancreas of rat from group 1 C (-) showing normal pancreatic acini and normal islets of Langerhans's.



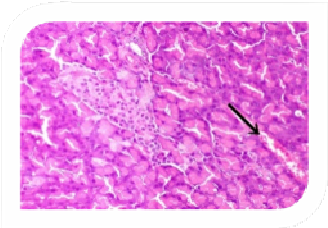
Picture (2): Photo of pancreas of rat from group 2 C (+) showing cystic dilatation of pancreatic duct (black arrow) and inflammatory cells infiltration (red arrow).



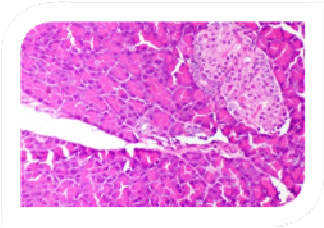
Picture (3): Photo of pancreas of rat from group 3 (2.5%) powder showing necrosis of some cells of islets of Langerhan's (black arrow) and vacuolation of some acinar epithelium (red arrow).



Picture (4): Photo of pancreas of rat from group 4 (5%) powder showing apparent normal pancreatic parenchyma.



Picture (7): Photo of pancreas of rat from group 5 (250 mg) extract showing slight congestion of pancreatic blood vessel (arrow) (H & E X 400).



Picture (8): Photo of pancreas of rat from group 6 (500 mg) extract showing no histopathological changes (H & E X 400).

Conclusion

Harmala seeds and extract have been shown to dramatically enhanced serum glucose levels ($P \leq 0.05$), improve HDL-c, blood parameters, and decrease liver and renal functions. This suggests that Harmala seeds could be added as daily drinks.

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تأثير بذور الحرمل على التغيرات البيولوجية والكيميائية الحيوية والنسيجية المرضية في الفئران المصابة بداء السكرى المستحث بالألوكسان

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الملخص العربي

يهدف البحث الحالي إلى التعرف على تأثير مسحوق ومستخلص بذور الحرمل على الفئران المصابة بمرض السكري المستحث الألوكسان. تم تقسيم ستة وثلاثين من الفئران البيضاء بطريقة عشوائية تتراوح أوزانها بين ١٤٠ .١٥٠ جرامًا إلى مجموعتين، كل منهما تتكون من ستة فئران. المجموعة (I): تم تقديمها كمجموعة ضابطة سالبة، والمجموعة (II): وتلقت المجموعة الثانية جرعة واحدة من مادة الألوكسان المسبب لمرض السكري (١٥٠ ملجم/كجم من وزن الجسم). تم بعد ذلك تقسيم الفئران المصابة بمرض السكري في المجموعة الثانية إلى ٥ مجموعات فرعية بشكل عشوائي: المجموعة ٢، المجموعة التي تعمل كمجموعة ضابطة موجبة، والتي أعطيت فقط النظام الغذائي القياسي؛ المجموعات ٣- ٦، التي تلقت النظام الغذائي القياسي بالإضافة إلى مسحوق بذور الحرمل بنسبة ٢،٥، ٥٪، والمستخلص ٢٥٠، ٥٠٠ مجم / كجم، لمدة ٤ أسابيع. تم قياس بعض التحاليل الكيميائية الحيوية؛ بعد ٢٨ يومًا من انتهاء التجربة. أظهر نتائج البيانات أن المجموعات المعاملة بالبذور انخفضت نسبة السكر والبروتين الدهني منخفض الكثافة والبروتين الدهني منخفض الكثافة جدا في الدم وزيادة البروتين الدهني مرتفع الكثافة (17e±1.71.29,101.70e±3.12 , 34.15.90b±4) ومؤشرات الدم مقارنة بالمجموعة الضابطة الموجبة. بالإضافة إلى ذلك، عززت النباتات المدروسة وظائف الكلى والكبد ومؤشرات الدم. ,153.61c±2.5542 .90.73c±1) .154.52c±2.61) وفقًا للنتائج، أن مسحوق بذور الحرمل ومستخلصه يتكون من مجموعة متنوعة من المركبات الكيميائية النباتية النشطة التي قد تقلل أيضًا من الآثار الضارة وتحسن من مستوى السكر في الفئران التي تعاني من ارتفاع سكر الدم.

الكلمات المفتاحية: الكوليسترول الكلي، جلوكوز الدم، وظائف الكبد، وظائف الكلى، ألانين أمينوترانسفيراز، أسبارتات أمينوترانسفيراز، الفوسفاتاز القلوي

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